مدارس الإمارات الوطنية
Emirates National Schools

Grade: 6 | Term 1, Week 12 (16 Nov to 20 Nov) | TCD: Future Skills 6: Applied Technology Foundations | Session length: 2 hrs

T1 WEEK: 12	**Lesson 1: Photon AI: Sense, Think, Act**
Standard(s):	Technology: algorithms, inputs and outputs; Math: 6.G (angles and distance), 6.EE (variables); Science: systems and cause and effect
Literacy Strategies:	System diagram labels; flowchart symbol cards (start or end, process, decision); frame: Photon senses ... thinks ... and acts by ...
Assessment:	Labelled system diagram; flowchart and tested route; live quiz on the TCD website.
Lesson Objectives:	We are learning to describe a robot as a system and to program precise multi-step movements.
Success Criteria:	I can explain sense, think and act using Photon as an example. I can label Photon's sensors (inputs) and outputs on a system diagram. I can program a route with distances and turns at set angles. I can draw a flowchart for my algorithm and update it when I change the code.
Targeted Vocabulary	system, input, output, sensor, algorithm, flowchart, sequence, angle, distance encoder
Resources	Photon robots, charged, Tablets with the Photon coding app, Mission mats, Protractors and measuring tapes, System diagram sheets, Flowchart symbol cards Accompanying resource: Grade6_Week12_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade6_Week12_Worksheet and Grade6_Week12_Lesson_Plan.
National Identity Integration	Link to the UAE's AI strategy and robots used in Dubai, such as delivery robots and smart police stations, as real sense-think-act systems.
AI/Digital Learning	Photon robots and tablets with the Photon coding app; hook video; live quiz on the TCD website.
Direct Instruction: I Do	Explain robots as systems (slide 6). Show Photon's sensors and outputs and label the diagram together (slide 7). Model coding a precise move and turn (slide 8) and drawing a flowchart (slide 9).
Guided Practice: We Do	Pairs label the system diagram. Class codes a short square together and compares the result with the plan (slide 10).
Independent Practice: You Do	Pairs plan, code, test and adjust their route and keep the flowchart up to date (slide 12). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: Beginning / On / Above	**Beginning: Code a square route with 90 degree turns using the flowchart template.** **On: Code a route with at least three different turn angles.** **Above: Code a route that draws a regular hexagon and explain the angle calculation (360 divided by 6).**
Closure/Reflection	Pairs share how far off the target they were and what they adjusted. Exit ticket (slide 15). Robots back to charging.
Inclusion / EAL Support	☑ Visual Supports ☐ Chunked Instructions ☐ Reduced Language Load ☑ Model / Worked Example ☐ Pre-teach Vocabulary ☐ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☑ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☑ Scaffold / Sentence Starters

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G&T / Extension	☑ Extension / Challenge Task ☐ Real-World Application ☑ Higher-Order Questioning ☐ Curriculum Compacting ☑ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☐ Leadership / Peer Coaching ☐ Open-Ended Task ☐ Other: _____